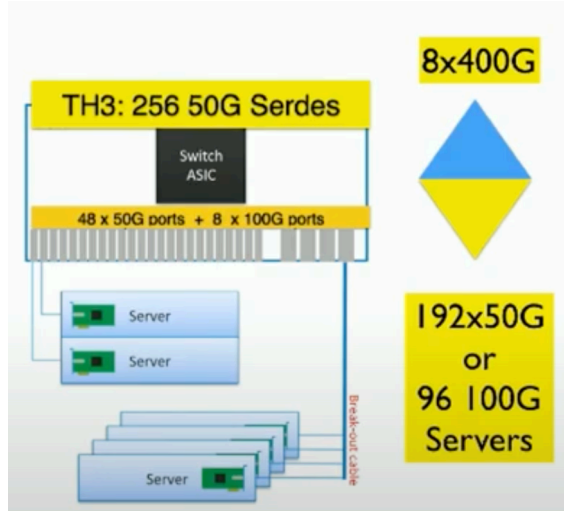




**Available Rack Bandwidth 1.228 Tbits/sec. (divide by 2 for TOR (redundant transmission on 2 switch ports) ~ 0.614 Tbits/sec**

Conclusion: there is 16 times as much 'Rack bandwidth' (East-West) available in the Transaction Fabrix as there is in Clos Architecture (compared to Arista) with the same serdes assumptions.

Trivial objections are straightforward to refute - Flat networks vs. Graph Networks. GVM enables one hop advantage. Arista's assumptions are for a flat network (minimum 2 hops with a TOR. Any application can talk to any other application. AVB's assumptions are *flawed*, and *self-serving*, to promote Arista's switches. All modern microservice applications are graphs, not fully connected A2A state management engines.

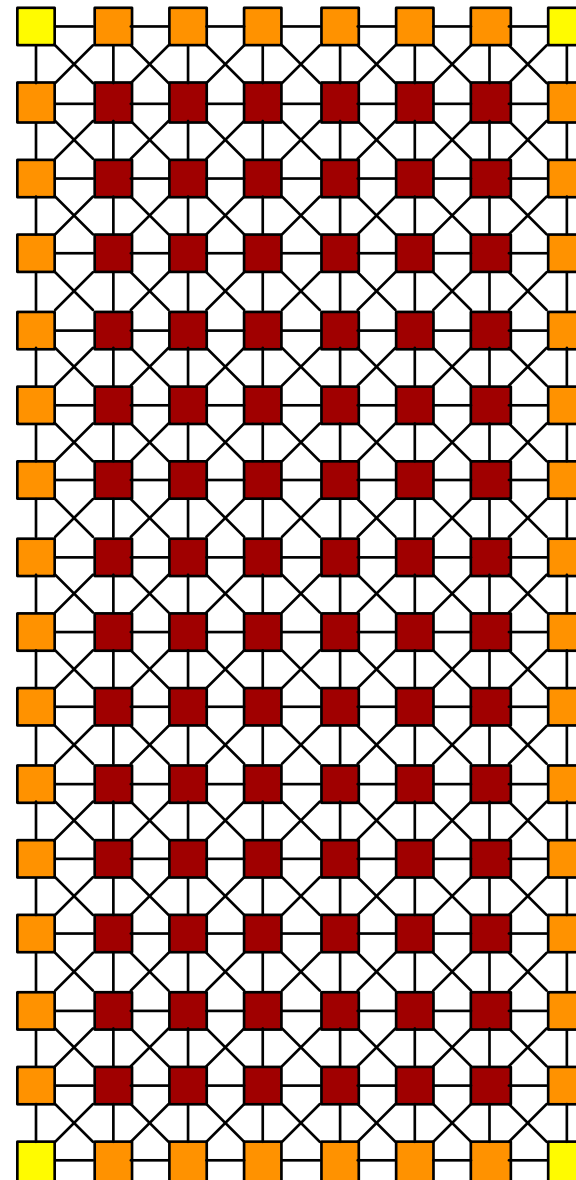
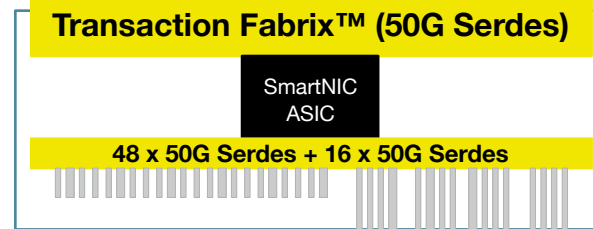
The above analysis is simplistic, it assumes a single TOR Switch, which is a Single Point of Failure (SPOF) and a bottleneck. Continue the argument: you need 2 TOR's (for redundancy), and you use two server ports (one connecting to each redundant TOR), The above analysis remains the same in cost ratios.

The underlying assumptions, we claim; is that the networking industry is in opposition to the distributed systems community in their optimizations. The bigger problem is that we need to serve the distributed systems community first; Even if the networking community is disrupted.

Other observations from AVB presentation: We copper cables for the downlink because it's the lowest cost.

From Jeff Tantsura: the cost of the cables is greater than the cost of the Switches. We are disrupting the cable industry. In reality, we are disrupting those that make the most margin from cables.

**This includes Arista**



**8 x 400G = 16 x 50G**

**96 x 100 G Servers**

(Total 96 Serdes)

No Stranded Ports

No Stranded Servers

No Stranded Power

No Stranded Slots

No break-out cables required!

**Total Rack Bandwidth = 9.830 Tbits/sec.**